

Mahesh Manikanta Kambhampati

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Summary

Proactive Security Analyst with hands-on experience in SIEM alert tuning, incident response, and endpoint detection across security operations environments supporting AWS and M365 cloud services. Skilled in threat detection, containment, eradication, and recovery aligned with NIST 800-61 incident response standards, with proficiency in vulnerability scanning, remediation tracking, and PCI audit evidence gathering. Experienced with Splunk, CrowdStrike Falcon, Microsoft Sentinel, Tenable Nessus, Qualys, Proofpoint, Snort, Suricata, Wireshark, and Python for alerting use case creation, automation, Threat Intel integration, and continuous monitoring. Detail oriented contributor experienced validating IPS/IDS alerts, reviewing firewall and network segmentation policies, and producing operational runbooks and metrics for security leadership.

Education

University of North Carolina

Master of Science, Cybersecurity

Aug 2023 – Dec 2024

Skills

Programming & Scripting: Java, Python, C, Bash

Web Development: HTML, CSS, JavaScript

Security Tools & Technologies: Splunk, Microsoft Sentinel, Cortex, CrowdStrike Falcon, Microsoft Defender, Nmap, KnowBe4, Tenable.io, Qualys, Nessus, Nikto, Burp Suite, OWASP Zap, Metasploit, Rapid7, Checkmarx, Proofpoint, DirBuster, John the Ripper, Wireshark, Snort, Suricata, FTK Imager, Autopsy, Ghidra, Okta, CyberArk, Active Directory, Entra ID, Purview, Symantec, Synk

Protocols & Networking: HTTP, HTTPS, FTP, SSH, Telnet, RDP, SMB, TCP/IP, OSI

Standards & Frameworks: NIST, ISO27001, PCI DSS, HITECH, HIPAA, MITRE ATT&CK

Operating Systems & Cloud: Kali Linux, Windows, Ubuntu, AWS, Azure

Work Experience

Advanced Energy

SOC Administrator

May 2026 – Present

Fort Collins, CO

- Tuned SIEM alerting use cases inside Splunk correlation searches, reducing false positive volume 35% and improving threat detection visibility across security operations.
- Investigated endpoint incidents using CrowdStrike Falcon real time response, performing containment and eradication on compromised hosts and documenting recovery actions for reporting.
- Supported vulnerability scanning processes with Tenable Nessus credentialed assessments, tracking remediation across 120 servers and coordinating patching timelines with IT teams.
- Validated Secure Email Gateway policies through Proofpoint targeted attack protection, blocking 200 phishing attempts monthly and strengthening threat prevention across M365 cloud environments.
- Automated log enrichment workflows with Python parsing scripts, integrating Threat Intel indicators into Splunk scheduled searches and scaling security operations monitoring efficiency.
- Produced incident response metrics using Splunk dashboard studio, summarizing containment timelines and escalation trends for IS leadership review aligned with NIST 800-61 standards.

ElevateX

Cybersecurity Engineer

Mar 2025 – May 2026

Charlotte, NC

- Operated EDR alert triage workflows using CrowdStrike Falcon detection dashboard, processing 30 events daily and classifying threats against established playbook procedures consistently.
- Examined IDS alert patterns through Suricata rule management console, correlating network analyzer findings with Threat Intel advisories and surfacing emerging threat trends promptly.
- Captured forensic artifacts using CrowdStrike Falcon host investigation, supporting senior engineers during security incidents from identification through eradication and recovery documentation.
- Contributed PCI compliance evidence by extracting audit logs from Splunk saved searches, organizing security event data to support annual PCI audit and IT general controls review.
- Tested endpoint product configurations within CrowdStrike Falcon prevention policies, verifying detection efficacy and confirming production readiness before deployment across enterprise environments.

- Facilitated analyst onboarding by demonstrating SIEM alerting workflows inside Splunk enterprise console, sharing alert classification practices and improving team triage accuracy 20%.

University of North Carolina

Jan 2024 – Dec 2024

Graduate Assistant

Charlotte, NC

- Monitored SIEM platform health across Splunk enterprise console, triaging alerts by severity and ensuring continuous monitoring resilience across endpoint and cloud security toolsets.
- Analyzed IPS alert data through Snort signature management, identifying suspicious network activity and escalating validated threats as tier escalation point for incident coordination.
- Executed vulnerability assessments with Qualys asset inventory, prioritizing findings against risk severity and tracking remediation follow up to reduce exploitable exposure 30%.
- Assisted use case creation by refining Microsoft Sentinel analytics rules, developing alerting automation for authentication anomalies and validating detection coverage against external threats.
- Reviewed firewall and proxy logs using Wireshark display filters, verifying network segmentation policies and flagging two configuration gaps per audit cycle for remediation planning.
- Drafted operational runbooks covering alert triage procedures inside Splunk report builder, standardizing support documentation for tier 1 and tier 2 analyst workflows consistently.

Certifications

- Google Cybersecurity Professional
- Certified Solutions Architect - Associate
- CompTIA Security+
- Junior Penetration Tester (eJPT)
- Merit Certificate and consolation prize in Cybersecurity Hackathon

Projects

Empowering Digital Parenting with Key Log Monitor | Python, Email Library, OS Library

- Built a Python based keylogger security solution with real time detection, automated email alerting, and proactive blocking, significantly strengthening endpoint monitoring and incident response capabilities.

Triple Layer Data Security | Java, Cryptographic Techniques

- Engineered a three layer Java encryption framework using advanced cryptographic algorithms, safeguarding sensitive data across storage and transmission channels, maximizing integrity and confidentiality.

Malware Analysis | CFF Explorer, Process Explorer, Ghidra, OllyDbg

- Conducted static & dynamic malware analysis, uncovering anti debugging techniques, network behaviors, and data exfiltration paths to enhance threat intelligence and incident response effectiveness

Forensic Evidence Investigation | FTK Imager, Autopsy, Belkasoft

- Conducted advanced digital forensic investigations using FTK Imager and Autopsy to extract critical evidence from file systems and volatile memory, compiling meticulous reports to support incident response and inform threat mitigation strategies.